# PAN: Part Attention Network Integrating Electromagnetic Characteristics for Interpretable SAR Vehicle Target Recognition

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***Abstract*— Machine learning methods for synthetic aperture radar (SAR) image automatic target recognition (ATR) can be divided into two main types: traditional methods and deep learning methods. The deep learning methods can learn the high-dimensional features of the target directly and usually obtain high target recognition accuracy. However, they lack full consideration of SAR targets' inherent characteristics, resulting in poor generalization and interpretation ability. Compared with deep learning methods, traditional methods can get more interpretable and stable results with model-based features. In order to take full advantage of these two kinds of methods, we propose the target part attention network (PAN) based on the attributed scattering center (ASC) model to integrate the electromagnetic characteristics with the deep learning framework. First, considering the importance of scattering structure for SAR ATR, we design a target part model based on the ASC model. Then, a novel part attention module based on the scaled dot-product attention mechanism is proposed, which directly associates the features of target parts with the classification results. Finally, we give the derivation method of the importance of each part, which is of great significance for practical application and the interpretation of SAR ATR. Experiments on the MSTAR dataset demonstrate the effectiveness of the proposed PAN. Compared with existing studies, it can achieve higher and more robust classification accuracy under different complex conditions. Furthermore, combined with the importance of parts, we constructed two effective interpretable analysis methods for deep learning network classification results.**

***Index Terms*— Attention mechanism, attributed scattering center (ASC) model, interpretable deep learning, synthetic aperture radar (SAR) target classification.**

## I. Introduction

SYNTHETIC aperture radar (SAR) can provide high-resolution and massive images independent of daylight, weather conditions, and cloud coverage [1]. It can monitor terrain and targets on the surface of the Earth via a global, reliable, and continuous way. SAR images have many applications in both military reconnaissance and civil observation areas [2], [3], [4]. Automatic target recognition (ATR) is an important part of SAR image intelligent interpretation, and machine learning (ML) techniques are widely used in this field.

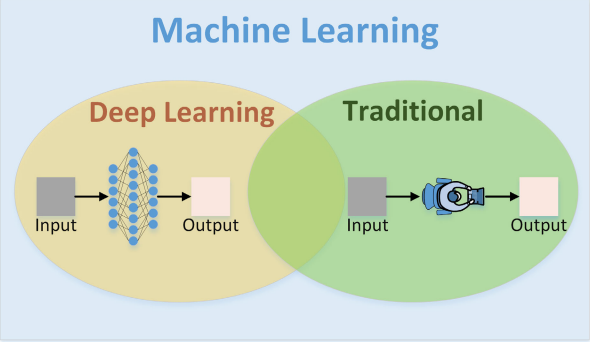



Fig. 1. Deep learning versus traditional ML method.

In the current data-intensive world, ML techniques are becoming more and more significant. They can be divided into two main subsets [5]: the deep learning method and the traditional method, as shown in Fig. 1. The former methods mainly use large data and rich computing resources to improve prediction performance, while traditional ML methods depend on handcrafted feature extraction and classifier design. In the SAR data area, ATR attempts to use ML methods that are specifically proposed for SAR data or come from the existing studies in the computer vision field to maximize target recognition performance [6].

For the traditional ML method, handcrafted feature extraction is required, and the widely used features in SAR target recognition include image domain features, transform domain features, electromagnetic scattering features, and so on. For typical image domain features, including the binary robust invariant scalable keypoints features [7], histogram of oriented gradients (HOG) features [8], and contour and shape features of targets [9], they closely reflect the characteristics of SAR images, but their classification performance is poor. For transform domain features, the principal component analysis, the linear discriminant analysis, the independent component analysis, and their variants have been broadly applied in

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SAR image feature extraction [10]. Although these methods are common and have good classification performance, they are mathematical analysis methods and poorly interpretable. In recent years, the traditional SAR ATR methods using electromagnetic scattering features have been a research hotspot. The electromagnetic scattering feature is a type of feature extracted from the raw data of an SAR target. Thus, it represents the inherent physical characteristics of the target, including material, structure, geometry information, and so on. The theoretical analysis demonstrates that the backscattering echo of the target can be modeled as the sum of the responses of individual attributed scattering centers (ASCs). The position and attribute information in the ASC model well reflect the structural characteristics of the local scattering points of the target. Many studies have shown the effectiveness of this model in SAR target feature extraction [11], [12], [13]. Although the overall recognition rate of the methods with traditional features is not outstanding, these methods have high stability in practical applications, especially those combining scattering characteristics, which have good explainability.

During the past decade, deep learning has made great progress in the field of computer vision and image processing with the improvement of device capabilities, including computing power, memory capacity, and image sensor resolution. As a conventional structure in computer vision, the convolutional neural network (CNN) [14] has the ability to extract the features of the target from the original images comprehensively and automatically with the convolution and pooling layers. Due to the outstanding performance of CNN in the field of visual image classification, many scholars have applied CNN models to SAR ATR. However, the network parameters of these models inherited from the computer vision field are huge, and the problem of overfitting is easy to occur on SAR images with a relatively small amount of data. To this end, researchers have made great efforts to optimize the existing network structure. In [15], a simple network model called all-convolutional networks (A-ConvNets) was proposed, and it used convolutional layers to replace all fully connected layers, in order to reduce the number of network parameters. Similarly, most recent SAR target recognition studies directly apply modules that have good performance in optical images to further optimize target recognition performance in the SAR field [16], [17], [18], [19]. These networks can improve classification accuracy, but they do not reflect the unique characteristics of SAR data, resulting in a lack of explanatory and generalization capabilities.

According to the existing research, deep learning introduces the end-to-end learning process, which automatically calculates the most significant and descriptive features for each target with a specific class. However, due to the well-known black box characteristics of deep learning methods, interpretability is also a major concern [20], [21]. Introducing the attention mechanism into deep networks is a common method to improve the interpretability of algorithms. This kind of method introduces an attention vector, gives different weights to the features of different hidden layers in the network, and learns those weights during the training process so that we can get the importance of each feature to model learning and achieve the effect of explaining the model. This mechanism has been widely developed and applied in natural language processing and computer vision [22], [23]. In view of the effectiveness of this mechanism in optical image recognition, many researchers have also applied the attention module to SAR ATR [24], [25]. However, these efforts focus more on the improvement of classification performance and do not analyze the interpretability of the algorithm. Therefore, for SAR ATR, the development of explainable deep learning techniques is still in the initial exploration stage.

Compared with traditional ML techniques, deep learning can usually achieve higher accuracy in target classification tasks. At the same time, using this method requires less expert analysis and can take advantage of the large data currently available. However, the end-to-end learning mode provides poor interpretability of the results, and the reliability of identification results in practical applications cannot be guaranteed. In addition, we find that the effective features learned from the deep neural network depend on the quality and quantity of the training dataset. Therefore, the generalization performance is poor when test images are different from the training set. In comparison to deep learning methods, traditional methods can solve some specific classification problems with simpler and faster technologies. The traditional ML algorithm has complete transparency and can adjust parameters under the guidance of experts to perform well in a wider range of tasks. Especially for SAR data, its unique electromagnetic characteristics can construct models and features with physical meaning. Therefore, traditional methods have better interpretability and robustness. Combining the advantages of the abovementioned two categories of methods, it is easy to find that incorporating the traditional model into the deep learning algorithm can complement their advantages and further improve the performance of target recognition for SAR images. Meanwhile, through the embedding of physical models, we can theoretically analyze the interpretability of deep learning for SAR ATR. As shown in the overlapping regions of the two methods in Fig. 1, fusion algorithms also become a current research hotspot [26], [27], [28].

### A. Related Works and Motivation

Based on the previous discussion, deep learning and traditional ML methods have their own advantages and disadvantages. Thus, fusion strategies are common solutions in this area and can be divided into two main frameworks, as shown in Fig. 2: decision-level fusion (DLF) and feature-level fusion (FLF) [29].

In this article, DLF means that the final result of the algorithm is the average value of the prediction results of the deep learning method and the traditional ML method. Feng et al. [30] calculated the classification probabilities based on the global and local characteristics of the target, respectively, and then obtained the final classification results through decision fusion. This work first used the ASC model to divide the target parts, then applied the deep learning method to compute the target global image and the part images classification results, and obtained the final classification results by averaging. However, DLF is mainly a numerical calculation of

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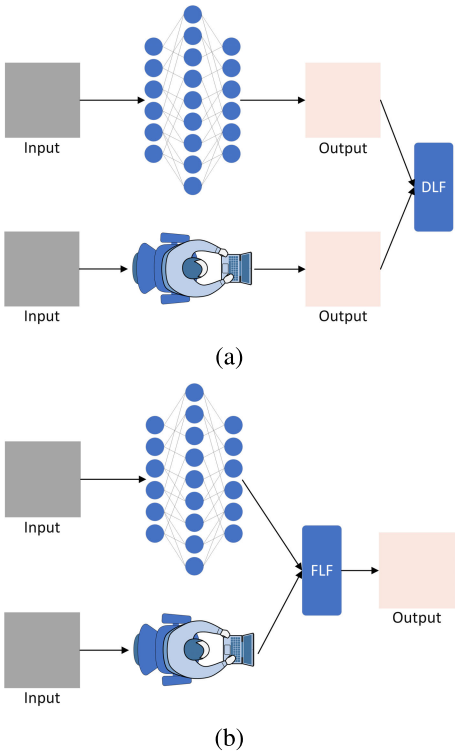



Fig. 2. Two main frameworks of fusion strategy: (a) DLF and (b) FLF.

the final classification probability and does not well represent the assistant effect of target electromagnetic characteristics on deep learning. Therefore, most of the current research is focused on FLF. FLF refers to combining traditional hand-crafted and deep-learning-based features. Then, the traditional ML classifier or the output network structure is used with the input of the fused features to obtain the final classification results. Zhang et al. [31] proposed a novel SAR ATR fusion framework. This work first extracted the features of the target from the deep learning model and the ASC model, respectively, then obtained the fused features through FLF, and, finally, calculated the classification results through the classifier. Li et al. [32] presented a multiscale SAR ATR network based on the component analysis. This article first obtained the component model of the target by clustering based on ASC parameters, then calculated the deep learning features of the component images and the target original image, respectively, and combined them. Choi et al. [33] integrated the automatic feature extraction ability of the deep learning network (DLN) with the fusion of target pixels and shadow information. Thus, it allowed the network to effectively utilize target characteristics and shadow signatures. Zhou et al. [34] mentioned that most of the CNN-based SAR target recognition methods only utilize the characteristics of SAR images but seldom utilize the unique electromagnetic scattering features of SAR targets. To solve this problem, they proposed a network that comprehensively used image features and ASC-related features to improve SAR ATR performance. Feng et al. [35], [36] creatively applied the concept of physics-guided deep learning to SAR ATR. Through the fusion of multiple network layer features, the ASC model was used to inform deep recognition networks.

In addition to vehicle ATR, FLF strategies are widely used in the detection and recognition of other targets such as aircraft and ships. Ai et al. [37] fused multiscale rotation-invariant Haar-like (MSRI-HL) feature with CNN for SAR ship detection. Guo et al. [38] proposed a scattering information enhanced attention pyramid network for SAR image aircraft detection. After the detection of the airport, they applied Harris—Laplace detector with scale and rotation invariance to model the scattering information of the target, and the information was used to enhance images for training and testing. Zhang et al. [39] pointed out that most of the existing in-depth learning studies focus on the modification of the algorithm structure and the loss function, resulting in limited performance improvement. To solve this problem, a novel network with the HOG feature fusion was proposed in their paper for SAR ship classification. Sun et al. [40] also extracted target scattering features and proposed keypoint loss based on the features to constrain aircraft recognition network training. The experimental results fully demonstrated the effectiveness of the network combined with scattering features in few-shot target recognition.

According to the relevant research of fusion algorithm, we find that FLF gets more attention and can obtain more robust classification performance. At the same time, the ASC model has received extensive attention in recent SAR ATR methods, which can describe the electromagnetic scattering features of the SAR target effectively. These inspire us to conduct the target recognition task by combining the electromagnetic scattering characteristics based on the ASC model and deep-learning features. However, for many existing studies, they only test the recognition performance of the proposed fusion algorithms. It is still difficult for them to interpret the classification results effectively, which is different from the original intention of the fusion algorithm research. Considering the wide application of the attention mechanism in optical images and the explainability of its principles, many studies have demonstrated the effectiveness of adding an attention mechanism to SAR ATR. The spatial attention module was introduced in [41] to fuse deep features of multiview SAR images. Cui et al. [42] proposed the spatial shuffle-group enhance attention module for SAR ship detection, which can be explained by the enhancement of advanced semantic information. Liu et al. [43] used the vision transformer (ViT) to extract global features of SAR images considering the transformer's capability to capture global characteristics. Wang et al. [44] also used a transformer to express the global characteristics of the target with local features. Although these efforts to introduce attention mechanisms make the network structure clearer, they are mainly used to improve network performance without making an explanatory analysis in this article. Therefore, this article focuses on how to achieve better fusion and make the algorithm interpretable.

### B. Contributions

In order to compensate for the deficiency of deep learning in understanding target characteristics, make deep learning less "black," and improve the robustness and explainability of the SAR ATR algorithm, a target part attention network

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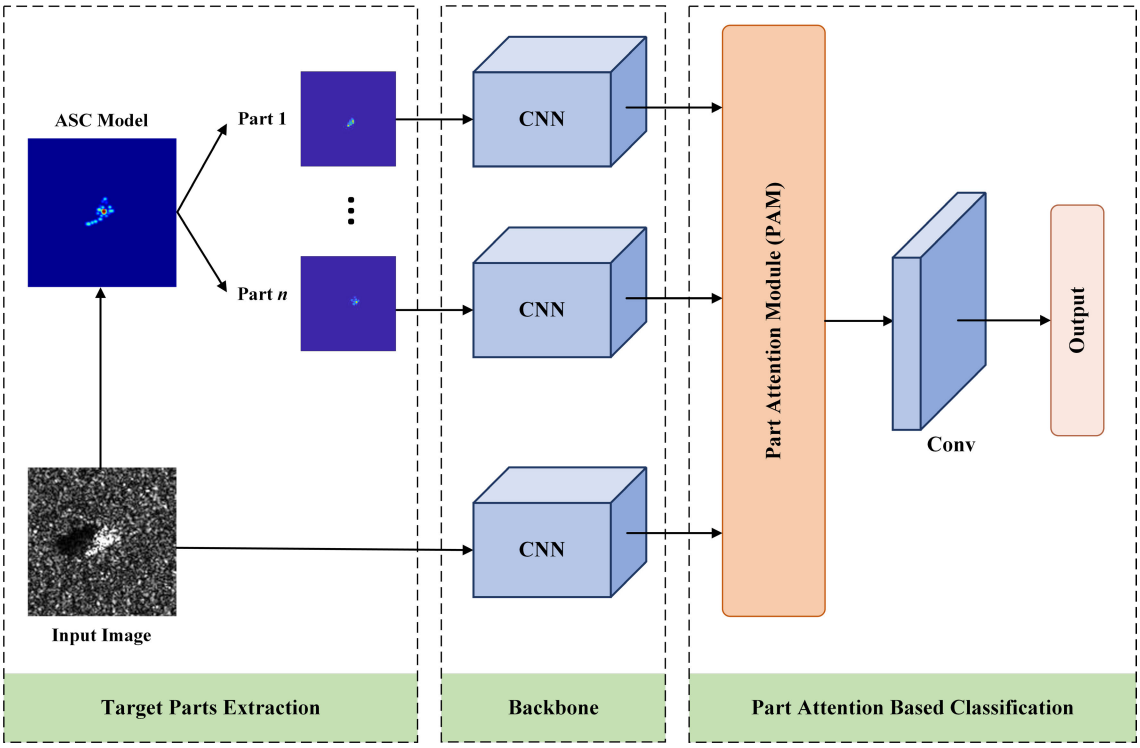



Fig. 3. Overall framework of the proposed PAN for SAR target recognition.

(PAN) integrating electromagnetic scattering characteristics is proposed. Considering the aforementioned problems, the main contributions of this article are listed as follows.

1) This article presents a novel part attention module (PAM) based on the ASC part model and attention mechanism to incorporate the electromagnetic characteristics of SAR targets into the DLN. This module enables the network to adaptively learn the impact of different target parts on classification tasks and directly associates the electromagnetic scattering characteristics of the target with the classification results.
2) Based on the PAM proposed, this article further presents an analysis method for the importance of parts to classification results, which is the first quantified analysis to our knowledge. Combining with the forward calculation method of the neural network, we directly deduce the importance of the part characteristics for the classification results, which is of great significance for practical application.
3) In terms of the interpretability of deep learning, we structure two analytical methods: part importance images and decision maps. We visualize the importance of each target's parts to the classification results under different experimental conditions and rationally interpret the classification results in combination with the actual scene and target characteristics. These methods provide new ideas and beneficial exploration for the explanation of the SAR ATR deep learning algorithm.

The remainder of this article is organized as follows. First, Section II presents the framework of PAN and the details of each module. Then, the experimental results of the proposed algorithm in comparison with others are given, and further interpretability analysis of the results is carried out in combination with the part importance defined in Section III. Finally, Section IV concludes this article.

## II. Part Attention Network

The overall structure of PAN is presented in Fig. 3. The structure consists of three main components: target parts' extraction based on the ASC model, CNN backbone for feature extraction, and part attention-based classification. Sections II-A–II-D describe the details of the above modules. At the same time, this section combines the network structure to analyze the calculation method of part importance.

### A. Parts' Extraction With ASC Model

As an extension of physical optics, the ASC model is developed based on the geometric theory of diffraction (GTD) [45], which assumes that the distributed object's radar backscattering in the high-frequency region can be approximated as the sum of the responses of each scattering center [46], as given in the following:

$$E(f,\varphi;\Theta)=\sum_{s=1}^{k}E_s(f,\varphi;\theta_s) \tag{1}$$

where the total number of the ASCs in the object is defined as $k$. $f$ and $\varphi$ denote the operating frequency and aspect angle of radar, respectively. For each individual ASC, $E_s$ and its corresponding physical parameter set $\theta_s$ are given as follows:

$$\begin{aligned}E_s(f,\varphi;\theta_s)&=A_s\cdot\left(j\frac{f}{f_c}\right)^{\alpha_s}\cdot\exp\left[-j\frac{4\pi f}{C}(x_s\cos\varphi+y_s\sin\varphi)\right]\\&\quad\cdot\operatorname{sinc}\left(\frac{2\pi f}{C}L_s\sin(\varphi-\bar{\varphi}_s)\right)\cdot\exp(-2\pi f\gamma_s\sin\varphi)\end{aligned} \tag{2}$$

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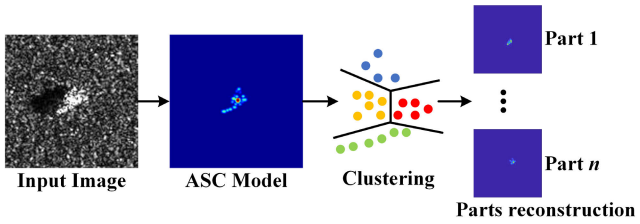



Fig. 4. Proposed target ASC parts' extraction process.

where $f_c$ is the radar central frequency and $C$ means the propagation speed of light. $[A_s, \alpha_s, x_s, y_s, L_s, \bar{\varphi}_s, \gamma_s]$ together constitute the parameter set of the $s$th scattering center, where $A_s$ is the relative amplitude and $\alpha_s$ reflects the frequency and geometry dependence. $(x_s, y_s)$ are the spatial coordinates, $L_s$ is the length of the distributed scattering center, and $\bar{\varphi}_s$ represents the orientation angle. $\gamma_s$ is the aspect dependence. These seven parameters determine the attribute information of each scattering center, especially $\alpha_s$ and $L_s$, which denotes the basic shape of the scattering center. The above parameters can be predicted by multiple optimization algorithms, and in this work, the genetic algorithm (GA) [47] is applied to predict the parameters.

In [30], we proposed the ASC model-based SAR target part images' calculation method, as shown in Fig. 4. First, the GA algorithm is used to predict seven parameters for each scattering center in the ASC model. Then, the K-means clustering algorithm is applied to divide the original ASC into multiple sets based on the parameters. After that, we calculate the reconstructed image for each scattering center cluster, as illustrated in (3). $E_R$ represents the data reconstructed by the extracted ASCs in the frequency domain according to the prediction parameter set. $\Theta_R = \{\theta_{\mathrm{Rs}}\}(s = 1, 2, \ldots, p)$ is the predicted parameter set, and $p$ denotes the number of scattering centers in a cluster. Finally, the 2-D inverse Fourier transform is performed on $E_R$ to obtain the reconstructed part image. Through this process, we can get the target ASC parts' images

$$E_R(f, \varphi; \Theta_R) = \sum_{s=1}^{p} E_{\mathrm{Rs}}(f, \varphi; \theta_{\mathrm{Rs}}). \tag{3}$$

Parts' extraction results of vehicle targets based on the K-means clustering algorithm with the parts number set to 5 and their corresponding target area images are shown in Fig. 5. Different colors in the figures represent different parts after clustering. It can be seen that the target parts extracted by this method cover the main target region of the SAR image, except for some areas with low echo amplitude. However, from the parts' distribution, we can see that the results of the K-means clustering method focus more on the location parameters of the scattering centers, that is, $(x_s, y_s)$. This feature results in a more homogeneous division of the target into parts by location. Thus, there is a difference between the results of such parts' extraction and our perception of the target composition.

In the existing research, there is also a method to extract target components by clustering scattering centers with two parameters $(\alpha_i, L_i)$ [32]. Different structures of targets can be divided by different combinations of these two parameters.

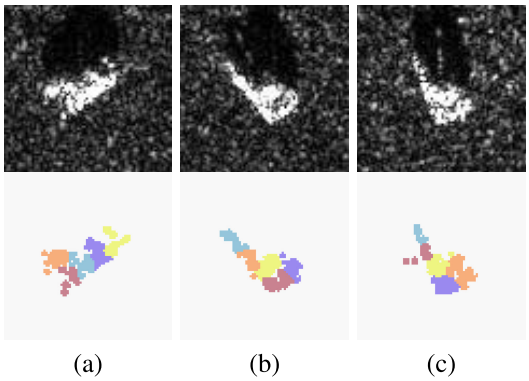



Fig. 5. Parts' extraction results of vehicle targets based on the K-means clustering algorithm. The first row represents the original SAR image of the target, and the second row represents the parts' extraction results of the corresponding target. Different colors only represent different parts.

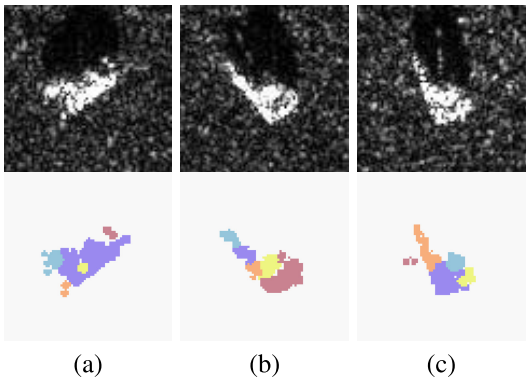



Fig. 6. Parts' extraction results of vehicle targets based on the SC algorithm. The first row represents the original SAR image of the target, and the second row represents the parts' extraction results of the corresponding target. Different colors only represent different parts.

Although this method can highlight the role of attribute information in the ASC model in target part extraction, it is still difficult to obtain a reasonable target part extraction result without considering the location and amplitude information of the scattering centers. After consulting more clustering algorithms, we find that the spectral clustering (SC) algorithm [48] can theoretically obtain more reasonable results for target partitioning. SC was shown to outperform other popular clustering algorithms, such as K-means, arguably due to its ability to handle nonconvex clusters [49]. We use the SC algorithm to cluster the seven parameters of the scattering centers and reconstruct the clustering results into the target parts' images. The parts' extraction results calculated using the same target data shown in Fig. 5 are shown in Fig. 6.

Comparing the part images, we can see that the two different clustering algorithms do have different preferences for the properties of the scattering centers. The results obtained by the SC algorithm tend to divide the vehicle target into a larger body part and some other smaller parts. Compared with the K-means algorithm, the SC algorithm can better reflect the detailed structure of the target. However, the target part model used in this article does not have a clear physical meaning, which is also a problem that cannot be solved at this stage. It more closely reflects the location and amplitude distribution of the target scattering point, and the combination of some parts may contain key structures that determine the

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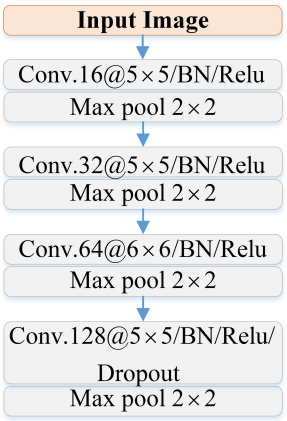



Fig. 7. Parameters and structure of the backbone network.

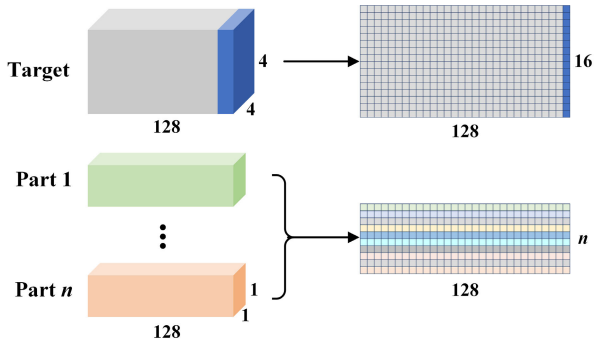



Fig. 8. Schematic of the feature dimension transformation process.

type of target. Therefore, the part model proposed can be understood as parts based on image semantics. Because the two clustering algorithms have different preferences, they are theoretically suitable for different classification scenarios. The two part extraction results will be further discussed in the experimental part in combination with the classification results under different conditions.

### B. Backbone Network for Feature Extraction

As illustrated in Fig. 3, after the target parts' extraction results are obtained, the features of the target data and the parts' images are extracted through the backbone network with the CNN structure. The parameters and structure of the backbone network are shown in Fig. 7.

The network structure is full convolution, inspired by A-ConvNets [15]. The backbone consists of four trainable convolutional layers with 16, 32, 64, and 128 channels, and their convolution kernels sizes are $5 \times 5$ and $6 \times 6$, respectively, as illustrated in the figure. After each layer, batch normalization (BN) and activation function with the rectified linear unit (ReLU) are operated. We add a max-pooling layer for downsampling after each convolutional layer with kernel size set to $2 \times 2$ and stride size set to 2. The dropout regularization technique is used in the last layer to set the output of each hidden unit to zero with a probability of 0.5.

The size of the input target image is $128 \times 128$. It is filtered by 16 convolution kernels of size $5 \times 5$ in the first convolution layer, resulting in 16 feature maps of size $124 \times 124$. After the first pooling layer, their sizes become $62 \times 62$. The outputs are then fed into the second convolution layer, and after the same pooling operation, the channel of the feature maps becomes 32, and the size of each map is $29 \times 29$. Similarly, after the third convolution and pooling layer, the feature maps become $64 \times 12 \times 12$. After the last convolution and pooling layer, the final output of the backbone network is $128 \times 4 \times 4$. Unlike the target image, we clipped the input parts' images to $88 \times 88$, considering the smaller part size. With the same network structure, the final output for each part channel is $128 \times 1 \times 1$.

### C. Part Attention-Based Classification

In part attention-based classification, three operations need to be carried out: first, the target and parts feature maps are unified through the feature dimension transformation; then, the correlation between parts and target is calculated through the PAM; and finally, the target feature dimension is restored through the inverse transformation of feature dimension for classification calculation.

*1) Feature Dimension Transformation:* According to the previous analysis, the dimensions of the target and parts feature maps calculated through the backbone network are $128 \times 4 \times 4$ and $128 \times 1 \times 1$, respectively. To calculate the relationship between the features of the part and the target, we need to transform the feature dimensions according to the process illustrated in Fig. 8.

For the target feature maps, we flatten the $4 \times 4$ size feature map in each channel into a 1-D feature vector. Then, the transformed feature map size is $128 \times 16$. For the parts feature maps, we convert the feature of each part into a $128 \times 1$ feature vector. Finally, the feature map size of $n$ parts is $128 \times n$.

*2) Part Attention Module:* PAM proposed in this article is inspired by the attention mechanism used in the transformer [50]: scaled dot-product attention. The attention mechanism is defined as mapping a query and a set of key-value pairs to the output. In the proposed PAM, we set the target feature map with $128 \times 16$ size as the query set, in which the 128-D feature vector of each channel is the query vector. The set of key-value pairs is obtained by the part's features, where the 128-D feature of each part corresponds to a key-value pair. Fig. 9 shows the part attention calculation process for a single-channel target feature vector in PAM.

$\{q_1, q_2, \ldots, q_{16}\}$ represent the query set calculated from the feature vectors of the 16 channels of the target. $\{(k_1, v_1), (k_2, v_2), \ldots, (k_n, v_n)\}$ represent the key-value pairs obtained by the parts features. Taking the part attention calculation of $q_1$ as an example, the weight of each part $\{\hat{\alpha}_{1,p} | p = 1, 2, \ldots, n\}$ is calculated as follows:

$$\alpha_{1,p} = \frac{q_1^{\text{Transpose}} \cdot k_p}{\sqrt{d_{q,k}}} \tag{4}$$

$$\hat{\alpha}_{1,p} = \text{Softmax}\left(\alpha_{1,p}\right). \tag{5}$$

$d_{q,k}$ means the dimension of the query and key vectors. We first compute the dot products of $q_1$ with all keys and divide the result by $\sqrt{d_{q,k}}$. Then, a Softmax function is applied to obtain the weights of the values. The final weighted

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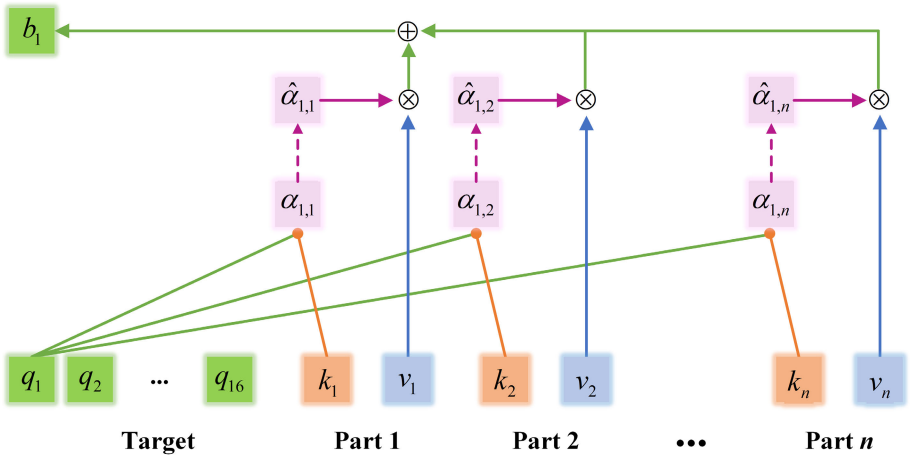



Fig. 9. Part attention calculation process for a single-channel target feature vector in PAM.

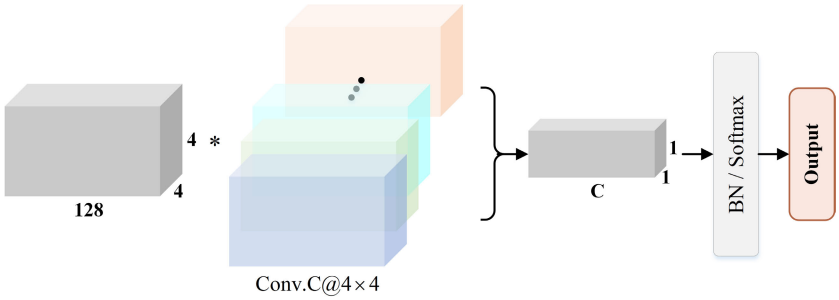



Fig. 10. Details of the classification head of the proposed network.

output is

$$b_1 = \sum_{p=1}^{n} \hat{\alpha}_{1,p} \cdot v_p. \tag{6}$$

Similarly, the parts' weighted output of other channels of target features can be calculated. Then, the final output of the PAM is $\{b_1, b_2, \ldots, b_{16}\}$ with the size of 128 × 16.

*3) Target Classification:* After obtaining the parts' weighted output, we inversely transform the output size to 128 × 4 × 4 according to the target feature dimension transformation method in Fig. 8. Through this step, we can use the convolution layer for target classification calculation instead of the full connection layer. Fig. 10 shows the details of the classification head of the proposed network. The output of PAM is filtered by $C$ convolution kernels of size 4 × 4. $C$ indicates the target category number. Then, BN and Softmax are applied to achieve the final classification output of the network.

### *D. Part Importance Analysis*

In order to analyze the role of different parts in target recognition and reflect the interpretability of the proposed algorithm, this section further studies the computational method of part importance based on the structure of PAN. Belloni et al. [51] use occlusion maps to study the location of features that contribute the most to correct classification in SAR target images. Inspired by their work, we used the method of part occlusion in previous studies [30] to express the importance of parts by the change of the class probability distribution vector after occlusion. However, in this method, the importance of the part only represents the part's role in identifying the single target image to which it belongs. The importance of different parts can only be compared in the same target image, and there is no significance of the horizontal comparison between different images. Therefore, in the above work, we can only roughly analyze the target parts of different categories that play a key identifying role in the classification algorithm by averaging the part importance maps of each category. By contrast, in the network proposed in this article, due to the use of PAM, the feature map in the network can be decomposed into the weighted sum of the feature inputs of each part of the target so that the output of the network can be directly quantified as the calculation results of each part. Thus, our algorithm can further interpret the target recognition network by comparing the importance of parts between different target images horizontally. Next, we will carefully introduce how to calculate the importance of target parts in combination with the algorithm in this article.

As shown in Fig. 10, we define the feature map with size of 128 × 4 × 4 as $\mathbf{F}$. It can be represented as a matrix of 4 × 4 128-D feature vectors, as shown in the following equation:

$$\mathbf{F} = \begin{bmatrix} f_{11} & f_{12} & f_{13} & f_{14} \\ f_{21} & f_{22} & f_{23} & f_{24} \\ f_{31} & f_{32} & f_{33} & f_{34} \\ f_{41} & f_{42} & f_{43} & f_{44} \end{bmatrix}. \tag{7}$$

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According to Section II-C, the feature map is obtained by feature dimension transformation. Therefore, for each element $f_{ij}$ in the feature map, the corresponding relationship with the PAM component weighted output is given as follows:

$$f_{ij} = b_{(i-1)\times 4+j}. \tag{8}$$

Let $t = (i-1) \times 4 + j$; we can represent $f_{ij}$ as the weighted calculation result of part features as follows:

$$f_{ij} = b_t = \sum_{p=1}^{n} \hat{\alpha}_{t,p} \cdot v_p. \tag{9}$$

For the convolution operation in Fig. 10, we denote each convolution kernel as $\{W^k | k = 1, 2, \ldots, C\}$, and the output of convolution operation is denoted as $\{\hat{f}^k | k = 1, 2, \ldots, C\}$. According to the calculation process of convolution, we can make the formula derivation as follows, where $w_{ij}^k$ represents the element in the convolution kernel $W^k$ corresponding to $f_{ij}$. It can be seen from the result of (10) that the output obtained by the convolution layer can be expressed in the form of the sum of convolution results of part features. $\hat{v}_{t,p}$ represents weighted part features

$$\begin{aligned}
\hat{f}^k &= \mathrm{Conv}_k(\mathbf{F}) \\
&= \sum_{j=1}^{4}\sum_{i=1}^{4} w_{ij}^k \cdot f_{ij} \\
&= \sum_{j=1}^{4}\sum_{i=1}^{4} w_{ij}^k \cdot \sum_{p=1}^{n} \hat{\alpha}_{t,p} \cdot v_p \\
&= \sum_{p=1}^{n}\sum_{j=1}^{4}\sum_{i=1}^{4} w_{ij}^k \cdot \hat{\alpha}_{t,p} \cdot v_p \\
&= \sum_{p=1}^{n}\sum_{j=1}^{4}\sum_{i=1}^{4} w_{ij}^k \cdot \hat{v}_{t,p} \\
&= \sum_{p=1}^{n} \mathrm{Conv}_k(\mathbf{V_p}).
\end{aligned} \tag{10}$$

After the convolution result is obtained, the final network output also needs BN and Softmax operations. For BN, the calculation method [52] is shown in (11). After the training, all the parameter values in the BN are fixed and the same for all the targets. Therefore, this scale normalization operation can be ignored when analyzing the importance of parts

$$y^k = \gamma \frac{\hat{f}^k - \mu^k}{\sigma^k} + \beta. \tag{11}$$

For Softmax, the calculation method [53] is given as follows. As illustrated in (12), the final output of the network can be expressed as the product of the parts convolution results. Thus, we can use $e^{\mathrm{Conv}_k(\mathbf{V_p})}$ to measure the role each part plays in classifying this target into class $k$. We take this value as the criterion for evaluating the importance of each part to the classification results, that is, the importance of each part to the classification result

$$y^k = \frac{e^{\hat{f}^k}}{\sum_{k=1}^{C} e^{\hat{f}^k}}$$

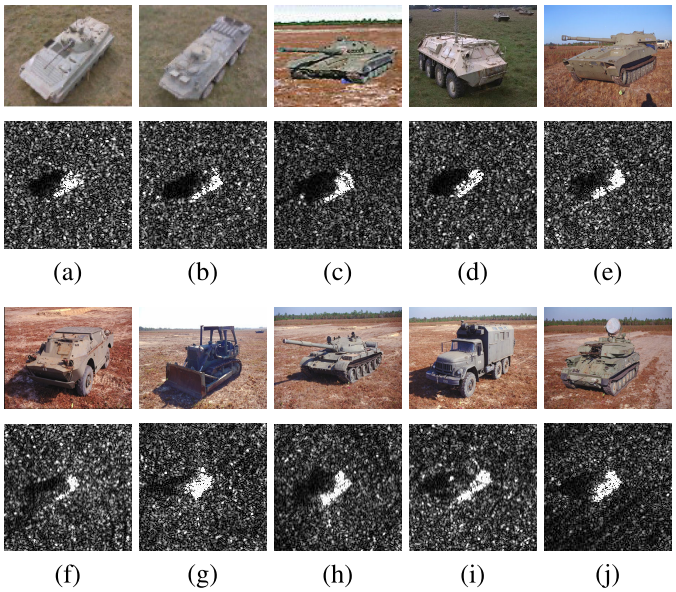



Fig. 11. Examples of ten types of ground targets. (a) BMP-2. (b) BTR-70. (c) T-72. (d) BTR-60. (e) 2S1. (f) BRDM-2. (g) D-7. (h) T-62. (i) ZIL-131. (j) ZSU-23/4.

$$= \frac{e^{\sum_{p=1}^{n} \mathrm{Conv}_k(\mathbf{V_p})}}{\sum_{k=1}^{C} e^{\hat{f}^k}}. \tag{12}$$

## III. Experiments

In this section, a number of experiments are conducted to validate the performance of the proposed PAN and illustrate the validity of part importance analysis. First, the dataset used in this article and different operation conditions are introduced. Then, we sequentially test the classification accuracy of the proposed algorithm with different experimental conditions compared with existing methods. Finally, the interpretability of the classification network is analyzed according to the proposed part importance images and decision maps. The MATLAB platform is used for ASC parts' extraction, and the Pytorch deep learning framework with NVIDIA GeForce RTX 2080Ti GPU is used to implement the experiments.

### A. Dataset

In this article, the Moving and Stationary Target Acquisition and Recognition (MSTAR) [54] dataset is utilized to validate the effectiveness of PAN. The resolution of the dataset is 0.3 m in the range direction and the cross-range direction. It contains ten different ground military targets with a full range of viewing angles: BMP-2, BTR-70, T-72, BTR-60, 2S1, BRDM-2, D-7, T-62, ZIL-131, and ZSU-23/4. Fig. 11 shows examples of the ten different ground targets, including optical images and their corresponding SAR images.

Standard operating conditions (SOCs) mean that the training and test images are obtained under similar imaging conditions, and their depression angles are slightly different, as shown in Table I. Extended operating conditions (EOCs) mean that the imaging conditions of the test images are different from those

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TABLE I
DETAILED DATA INFORMATION OF SOC

| Training | | | Test | | |
|---|---|---|---|---|---|
| Class | Depression | Num | Class | Depression | Num |
| BMP-2 | 17° | 233 | BMP-2 | 15° | 195 |
| BTR-70 | 17° | 233 | BTR-70 | 15° | 196 |
| BTR-60 | 17° | 256 | BTR-60 | 15° | 195 |
| T-72 | 17° | 232 | T-72 | 15° | 196 |
| T-62 | 17° | 299 | T-62 | 15° | 273 |
| 2S1 | 17° | 299 | 2S1 | 15° | 274 |
| BRDM-2 | 17° | 298 | BRDM-2 | 15° | 274 |
| D-7 | 17° | 299 | D-7 | 15° | 274 |
| ZIL-131 | 17° | 299 | ZIL-131 | 15° | 274 |
| ZSU-23/4 | 17° | 299 | ZSU-23/4 | 15° | 274 |

TABLE II
DETAILED DATA INFORMATION OF EOC1

| Training | | | Test | | |
|---|---|---|---|---|---|
| Class | Depression | Num | Class | Depression | Num |
| T-72 | 17° | 299 | T-72 | 30° | 288 |
| BRDM-2 | 17° | 298 | BRDM-2 | 30° | 287 |
| 2S1 | 17° | 299 | 2S1 | 30° | 288 |
| ZSU-23/4 | 17° | 299 | ZSU-23/4 | 30° | 288 |

TABLE III
DETAILED DATA INFORMATION OF EOC2/C

| Training | | | Test | | |
|---|---|---|---|---|---|
| Class/Series | Depression | Num | Class/Series | Depression | Num |
| BMP-2/9563 | 17° | 233 | T-72/A32 | 15°, 17° | 572 |
| BTR-70/C71 | 17° | 233 | T-72/A62 | 15°, 17° | 573 |
| T-72/132 | 17° | 232 | T-72/A63 | 15°, 17° | 573 |
| BRDM-2/E71 | 17° | 298 | T-72/A64 | 15°, 17° | 573 |
| | | | T-72/S7 | 15°, 17° | 419 |

TABLE IV
DETAILED DATA INFORMATION OF EOC2/V

| Training | | | Test | | |
|---|---|---|---|---|---|
| Class/Series | Depression | Num | Class/Series | Depression | Num |
| BMP-2/9563 | 17° | 233 | BMP-2/9566 | 15°, 17° | 428 |
| BTR-70/C71 | 17° | 233 | BMP-2/C21 | 15°, 17° | 429 |
| T-72/132 | 17° | 232 | T-72/A04 | 15°, 17° | 573 |
| BRDM-2/E71 | 17° | 298 | T-72/A05 | 15°, 17° | 573 |
| | | | T-72/A07 | 15°, 17° | 573 |
| | | | T-72/A10 | 15°, 17° | 567 |
| | | | T-72/812 | 15°, 17° | 426 |

in the training dataset, including variations in depression angle, target configuration, and version variants. EOC1 refers to large depression angle variation, as shown in Table II. EOC2/C represents configuration variation of vehicle targets, as shown in Table III. EOC2/V means target version variation, as shown in Table IV.

### B. Results of Target Recognition Under SOC/EOCs

In this section, we evaluate the target recognition performance of PAN with the MSTAR data under different operating conditions. First, according to the description of the part extraction method in Section II-A, we analyze the classification performance of the algorithm using different part extraction results. Then, the classification results of the algorithm presented in this article are compared with those of existing research methods.

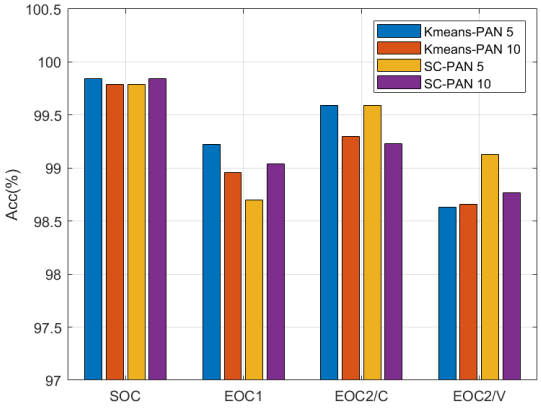



Fig. 12. Test classification accuracy of the proposed algorithms with different configurations.

*1) Performance With Different Part Extraction Results:* In Section II-A, two part extraction algorithms are mentioned. We name the proposed PAN with the input of these two kinds of part extraction results as Kmeans-PAN and SC-PAN, respectively. For each part extraction algorithm, we also compare the effect of different numbers of parts on the algorithm performance. Kmeans-PAN 5 means PAN using the K-means part extraction method, with the number of parts set to 5. Kmeans-PAN 10 means PAN using the K-means part extraction method, with the number of parts set to 10. The definition of SC-PAN is similar to the above. For all the algorithms with different configurations, we train with the MSTAR dataset under SOC and EOCs, and the final test classification accuracy is shown in Fig. 12.

From the results in the figure, we can see that, for different part extraction algorithms, there is no obvious difference in classification performance under SOC. However, through comprehensive comparison under EOC1, the proposed PAN with the parts input obtained by using K-means can obtain relatively higher classification accuracy. On the contrary, for EOC2, the parts obtained with the SC method can achieve significantly better classification performance. Combining the results of different parts' extraction in Section II-A, we can infer that the preferences of target parts are different in classification experiments under EOC1 and EOC2. The target parts obtained by K-means are more uniform and dispersed, while the results of SC contain one main large structure and several local small structures. The difference between EOC2 training data and test data is mainly reflected in some small structural changes, and the target body has high consistency, so the results obtained by SC are more suitable for this classification condition. This part of the comparative analysis will be further discussed in the following in combination with the proposed part importance analysis method. For different numbers of parts, we can see from the results in the figure that five parts have better performance than ten parts under most conditions. We think that this is mainly due to the small size of the vehicle target,

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TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT NETWORKS UNDER SOC AND EOCS

| Method | Accuracy (%) | | | |
|---|---|---|---|---|
| | SOC | EOC1 | EOC2/C | EOC2/V |
| ACN [15] | 99.13 | 96.12 | 98.93 | 98.60 |
| M-Net [55] | 99.71 | 97.48 | 99.67 | 98.74 |
| SCN [56] | 99.79 | 98.78 | 99.80 | 99.35 |
| FEC [31] | 99.59 | 99.19 | - | 98.48 |
| MKSFF-CNN [57] | 97.44 | 82.97 | - | - |
| IFTS-Net [33] | 98.90 | 96.86 | 98.28 | 95.46 |
| SDF-Net [58] | 99.58 | 97.16 | - | 98.76 |
| MS-CVNets [59] | 99.79 | 96.26 | - | - |
| Kmeans-PAN | **99.84** | **99.22** | 99.59 | 98.63 |
| SC-PAN | 99.79 | 98.70 | **99.59** | **99.13** |

TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT NETWORKS UNDER SOC AND EOCS

| Method | Accuracy (%) | | | |
|---|---|---|---|---|
| | SOC | EOC1 | EOC2/C | EOC2/V |
| MACN | 99.46 | 97.05 | 99.56 | 98.09 |
| Kmeans-PAN | **99.84** | **99.22** | 99.59 | 98.63 |
| SC-PAN | 99.79 | 98.70 | **99.59** | **99.13** |

which can be divided into five parts to ensure that each part has a certain structure. Setting the number of parts to 10 will make the target partition too fine, close to a single scattering center, and will not improve the algorithm's performance. Therefore, in the subsequent experimental analysis, five parts are used as input for PAN.

*2) Performance Compared With Other Methods:* To further illustrate the performance of the proposed algorithm, we compare the PAN with different methods in the existing literature. A-ConvNets (ACN) [15] is a classical deep learning algorithm in the field of SAR target recognition. M-Net [55] adds an information recorder to remember and store samples' spatial features, which has achieved relatively high classification accuracy in recent years. The SAR capsule network (SCN) [56] embeds Hinton's capsule network structure in CNN, which also achieves better performance. In addition, we compare the performance of our algorithm with many fusion algorithms: FEC [31], MKSFF-CNN [57], IFTS-Net [33], SDF-Net [58], and MS-CVNets [59]. Among these algorithms, FEC and SDF-Net are also methods that fuse ASC model-related features with deep learning features. FEC obtains the statistical features of target ASC parameters through the bag of visual words model, while SDF-Net further calculates the target features based on ASC parameters. The classification results of different algorithms under SOC and EOCs are presented in Table V.

From the results in the table, we can see that, for each experimental condition, the classification performance of our algorithm has basically reached the leading level. For the vast majority of existing algorithms, they have excellent performance mainly under individual experimental conditions and seldom adapt to all experimental conditions. Especially by comparing other fusion methods with our algorithm, we can see that the algorithm in this article can reach a higher level under both SOC and EOCs, which indicates that the proposed PAN has a better comprehensive classification performance. In addition, we find that most of the algorithms are prone to overfitting when they are experimented with EOC2/C training and test data directly. At the beginning of the training process of these algorithms, it is easy to obtain model parameters with high classification accuracy, which makes the classification performance meaningless. Considering the consistency of training data in EOC2, our algorithm calculates the test data of EOC2/C and EOC2/V using the same model parameters to obtain the final classification accuracy. Under such experimental settings, our algorithm can obtain higher classification accuracy in EOC2, which further illustrates that our algorithm has better robustness.

In addition to the above comparisons, we also conduct ablation experiments to verify the effect of the proposed PAM. From Fig. 3, we can see that, if part input and PAM are removed, the target original image is extracted directly from the backbone network, and the classification results are obtained through the full convolution layer. This process is similar to ACN, but, as described in Section II-B, some modifications have been made to the parameters of the network. For a network without a PAM, we define it as a modified ACN (MACN). Table VI presents the classification accuracy of these algorithms under SOC and EOCs.

It is not difficult to see from the table above that the performance of the algorithm is improved when the attention of the parts is combined. Especially under EOCs, the classification accuracy of PAN is significantly higher than that of MACN. Due to the complexity of the experimental conditions, this comparison further demonstrates the robustness of the proposed algorithm under depression angle and target version changes combined with part attention.

### C. Results of Target Recognition Under Complex Conditions

Since there are many studies on MSTAR target classification under SOC/EOCs at this stage, as shown in Table V, those algorithms can achieve good classification performance, and the advantages of the algorithm mentioned in this article are not obvious. In order to further analyze the algorithm performance, this section constructs more difficult experimental conditions based on SOC and EOC data to compare the performance of different algorithms.

*1) Series Variation Condition:* First, the proposed method is trained and tested under series variation condition (SVC), which means the training data and test data of the same class target are different in series number. The detailed data information of SVC is shown in Table VII. For BMP-2 and T-72, only one serial number is used in the training process, while all serial numbers' data are used during testing. With the same experimental data, besides the classical deep learning algorithms (DLN), we compare our method with other algorithms that also incorporate ASC models into the deep learning network (ASC + DLN), including ASC-Map [34], CA-MCNN [32], ASC-GCN [26], and GLFN [30]. Table VIII presents the classification accuracy of these algorithms under

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TABLE VII
DETAILED DATA INFORMATION OF SVC

| Training | | | Test | | |
|---|---|---|---|---|---|
| **Class/Series** | **Depression** | **Num** | **Class/Series** | **Depression** | **Num** |
| BMP-2/9563 | 17° | 233 | BMP-2/9563 | 15° | 195 |
| | | | BMP-2/9566 | 15° | 196 |
| | | | BMP-2/C21 | 15° | 196 |
| BTR-70/C71 | 17° | 233 | BTR-70/C71 | 15° | 196 |
| | | | T-72/132 | 15° | 196 |
| T-72/132 | 17° | 232 | T-72/812 | 15° | 195 |
| | | | T-72/s7 | 15° | 191 |

TABLE VIII
CLASSIFICATION ACCURACY OF DIFFERENT NETWORKS UNDER SVC

| **Method** | | **Accuracy (%)** |
|---|---|---|
| **DLN** | ACN [15] | 93.85 |
| | VGG [60] | 92.89 |
| **ASC+DLN** | ASC-Map [34] | 97.95 |
| | CA-MCNN [32] | 98.61 |
| | ASC-GCN [26] | 98.75 |
| | GLFN [30] | 98.46 |
| **Proposed** | MACN | 98.01 |
| | Kmeans-PAN | 98.75 |
| | SC-PAN | **99.19** |

TABLE IX
DETAILED ACCURACY OF DIFFERENT NETWORKS UNDER SVC

| **Method** | **Accuracy (%)** | | | |
|---|---|---|---|---|
| | **BMP2** | **BTR70** | **T72** | **Total** |
| ACN | 91.99 | 98.98 | 93.99 | 93.85 |
| CA-MCNN | 98.13 | 99.49 | 98.89 | 98.61 |
| ASC-GCN | 98.47 | 99.49 | 98.80 | 98.75 |
| MACN | 96.25 | 98.98 | 99.48 | 98.02 |
| Kmeans-PAN | **98.81** | 1 | 98.28 | 98.75 |
| SC-PAN | 98.46 | **1** | **99.66** | **99.19** |

SVC, and detailed accuracy of each type of target is shown in Table IX.

From Table VIII, it can be found that the classification accuracy of the algorithms combined with the ASC model is generally higher than that of the classical deep learning algorithms. At the same time, the SC-PAN proposed in this article achieves the optimal classification performance. By comparing the results in Table IX, we can see that the PAN achieves the highest classification accuracy on each target type, which fully demonstrates that our algorithm still has the best performance under SVC. In addition, by analyzing the differences between K-means and SC, we can draw a consistent conclusion with EOC2 that the target part result obtained by SC is more suitable for learning the changes in the target detailed structure.

*2) Fine-Grained Condition:* Besides, the proposed method is also trained and tested under fine-grained conditions (FGCs) to identify different subclasses within the same category of targets. The detailed data information of FGC is shown in Table X, BMP-2 has two subclasses, and T-72 has five subclasses. Tables XI–XIII present the classification accuracy and the confusion matrix of MACN, Kmeans-PAN, and SC-PAN under FGC.

TABLE X
DETAILED DATA INFORMATION OF FGC

| Training | | | Test | | |
|---|---|---|---|---|---|
| **Class/Series** | **Depression** | **Num** | **Class/Series** | **Depression** | **Num** |
| BMP-2/9566 | 17° | 232 | BMP-2/9566 | 15° | 196 |
| BMP-2/C21 | 17° | 233 | BMP-2/C21 | 15° | 196 |
| T-72/A04 | 17° | 299 | T-72/A04 | 15° | 274 |
| T-72/A05 | 17° | 299 | T-72/A05 | 15° | 274 |
| T-72/A07 | 17° | 299 | T-72/A07 | 15° | 274 |
| T-72/A10 | 17° | 296 | T-72/A10 | 15° | 271 |
| T-72/812 | 17° | 231 | T-72/812 | 15° | 195 |

TABLE XI
CLASSIFICATION ACCURACY AND CONFUSION MATRIX OF MACN UNDER FGC

| **Seies** | **9566** | **C21** | **A04** | **A05** | **A07** | **A10** | **812** | **Accuracy** |
|---|---|---|---|---|---|---|---|---|
| **9566** | **186** | 9 | 0 | 0 | 0 | 0 | 1 | 94.90% |
| **C21** | 8 | **186** | 0 | 0 | 1 | 0 | 1 | 94.90% |
| **A04** | 0 | 1 | **264** | 0 | 1 | 2 | 6 | 96.35% |
| **A05** | 0 | 0 | 0 | **272** | 0 | 2 | 0 | 99.27% |
| **A07** | 0 | 0 | 1 | 3 | **262** | 7 | 1 | 95.62% |
| **A10** | 0 | 0 | 0 | 0 | 0 | **271** | 0 | 100.00% |
| **812** | 0 | 0 | 1 | 0 | 0 | 0 | **194** | 99.49% |
| **Accuracy** | **97.32%** | | | | | | | |

TABLE XII
CLASSIFICATION ACCURACY AND CONFUSION MATRIX OF KMEANS-PAN UNDER FGC

| **Seies** | **9566** | **C21** | **A04** | **A05** | **A07** | **A10** | **812** | **Accuracy** |
|---|---|---|---|---|---|---|---|---|
| **9566** | **186** | 10 | 0 | 0 | 0 | 0 | 0 | 94.90% |
| **C21** | 1 | **195** | 0 | 0 | 0 | 0 | 0 | 99.49% |
| **A04** | 0 | 0 | **266** | 0 | 1 | 3 | 4 | 97.08% |
| **A05** | 0 | 0 | 1 | **272** | 0 | 1 | 0 | 99.27% |
| **A07** | 0 | 0 | 0 | 0 | **272** | 2 | 0 | 99.27% |
| **A10** | 0 | 0 | 0 | 0 | 0 | **271** | 0 | 100.00% |
| **812** | 0 | 0 | 1 | 0 | 0 | 0 | **194** | 99.49% |
| **Accuracy** | **98.57%** | | | | | | | |

TABLE XIII
CLASSIFICATION ACCURACY AND CONFUSION MATRIX OF SC-PAN UNDER FGC

| **Seies** | **9566** | **C21** | **A04** | **A05** | **A07** | **A10** | **812** | **Accuracy** |
|---|---|---|---|---|---|---|---|---|
| **9566** | **191** | 4 | 0 | 0 | 0 | 0 | 1 | 97.45% |
| **C21** | 2 | **194** | 0 | 0 | 0 | 0 | 0 | 98.98% |
| **A04** | 0 | 0 | **270** | 0 | 0 | 3 | 1 | 98.54% |
| **A05** | 0 | 0 | 0 | **273** | 0 | 1 | 0 | 99.64% |
| **A07** | 0 | 0 | 0 | 2 | **269** | 3 | 0 | 98.18% |
| **A10** | 0 | 0 | 0 | 0 | 0 | **271** | 0 | 100.00% |
| **812** | 0 | 0 | 0 | 0 | 0 | 0 | **195** | 100.00% |
| **Accuracy** | **98.98%** | | | | | | | |

Comparing the results in the three tables, it can be found that the PAN algorithm can achieve significantly better classification performance than the deep learning algorithm by fusing the part features based on the ASC model. At the same

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TABLE XIV
CALCULATION RESULTS OF PART IMPORTANCE FOR BMP-2 TARGETS

| Target | part1 | part2 | part3 | part4 | part5 |
|---|---|---|---|---|---|
| 1 | 0.7403 | 0.9504 | 0.9018 | 0.9198 | 1 |
| 2 | 0.6006 | 0.7919 | 0.9055 | 0.9378 | 1 |

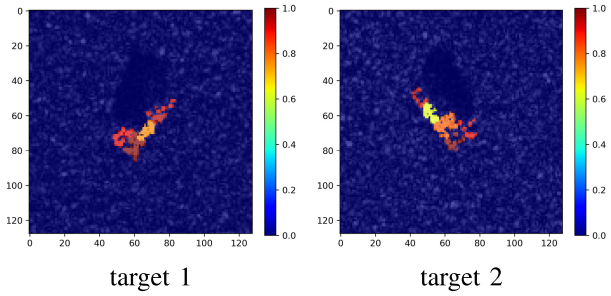



Fig. 13. Part importance maps of BMP-2 targets.

time, the proposed methods not only improve the recognition accuracy of different serial number targets but also reduce the number of samples misclassified between categories. In addition, SC-PAN can obtain higher accuracy than Kmeans-PAN, which further validates the previous conclusion that target parts based on SC are more suitable for classification with slight differences between targets.

### D. Part Importance Images of Experimental Results

In order to further analyze the role of target parts in classification, this section conducts some interpretable analysis of the proposed algorithm and experimental results. According to the experimental results in Section III-C, we can find that the part extraction results based on different algorithms are suitable for different experimental conditions. Therefore, we combined the part importance calculation method based on the part attention mechanism described in Section II-D to construct part importance images, which visually analyze different experimental results.

*1) Results of SOC Data:* As described in Section II-D, we first calculate the importance of each part of the target to the classification results: $e^{\mathrm{Conv}_k(\mathbf{V_p})}$, where $\mathrm{Conv}_k$ is the convolution layer of the corresponding target class channel in the trained network. Then, we assign corresponding values to the target parts to get the part importance maps for each target. Taking BMP-2 as an example, Table XIV shows the calculation results of part importance for some targets. The results are visualized by multiplying them with the corresponding parts pixels in the original images, as shown in Fig. 13. Comparing the results of the two targets, it can be found that, because the parts are divided for each target and there is no correspondence between the parts of different targets, it is meaningless to analyze from the value of importance. In Fig. 13, two targets with different azimuth angles have high consistency in the distribution of part importance, which also shows the correctness of the calculation method proposed in this article.

Also, considering inconsistent target parts, in order to analyze the part importance of each category target, only the average part importance map can be calculated by overlaying them all. In this process, all target images with different azimuths in the same category need to be rotated to the same azimuth, which is called angle normalization in this article. The average part importance images are represented as thermodynamic diagrams, and the redder the area, the more important the area is for the classification results. The average part importance images of MSTAR vehicle targets of Kmeans-PAN and SC-PAN under SOC with angle normalization are shown in Figs. 14 and 15, respectively.

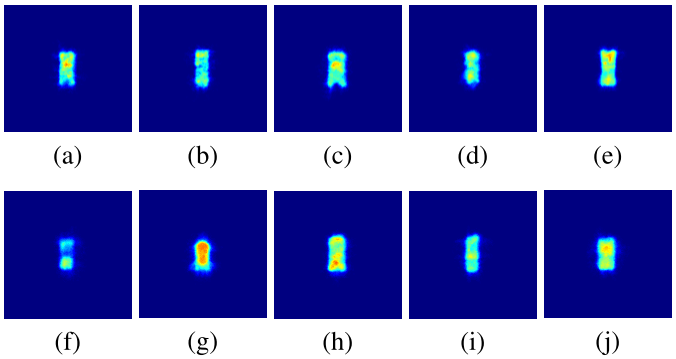



Fig. 14. Average part importance images of MSTAR vehicle targets of Kmeans-PAN under SOC with angle normalization. (a) BMP-2. (b) BTR-70. (c) T-72. (d) BTR-60. (e) 2S1. (f) BRDM-2. (g) D-7. (h) T-62. (i) ZIL-131. (j) ZSU-23/4.

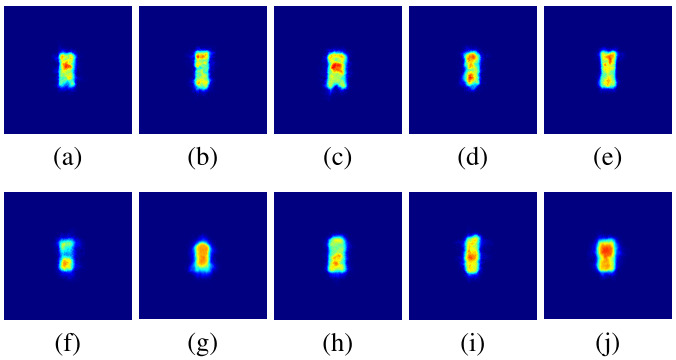



Fig. 15. Average part importance images of MSTAR vehicle targets of SC-PAN under SOC with angle normalization. (a) BMP-2. (b) BTR-70. (c) T-72. (d) BTR-60. (e) 2S1. (f) BRDM-2. (g) D-7. (h) T-62. (i) ZIL-131. (j) ZSU-23/4.

According to the experimental results in Section III-B, the classification performance under SOC obtained by using the two different part segmentation results for network training is similar. At the same time, we can also see from the figures that the average part importance images of different targets are relatively consistent under the two algorithms. This shows that the part importance calculation method based on PAN proposed in this article can reflect the role of different target parts in the classification process. In addition, the results of this article are very similar to those part importance maps calculated by the part occlusion method in our previous work [30]. This further illustrates the rationality of the proposed method for calculating the importance of parts. At the same time, it fully demonstrates that the target parts play a highly consistent role even in different classification algorithms.

*2) Results of EOC1 Data:* Using the same calculation method, the average part importance images of MSTAR

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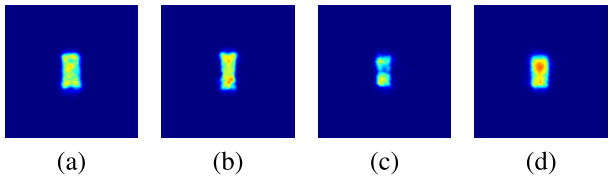



Fig. 16. Average part importance images of MSTAR vehicle targets of Kmeans-PAN under EOC1 with angle normalization. (a) T-72. (b) 2S1. (c) BRDM-2. (d) ZSU-23/4.

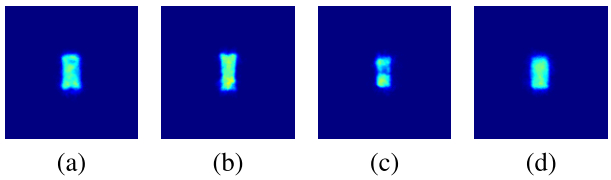



Fig. 17. Average part importance images of MSTAR vehicle targets of SC-PAN under EOC1 with angle normalization. (a) T-72. (b) 2S1. (c) BRDM-2. (d) ZSU-23/4.

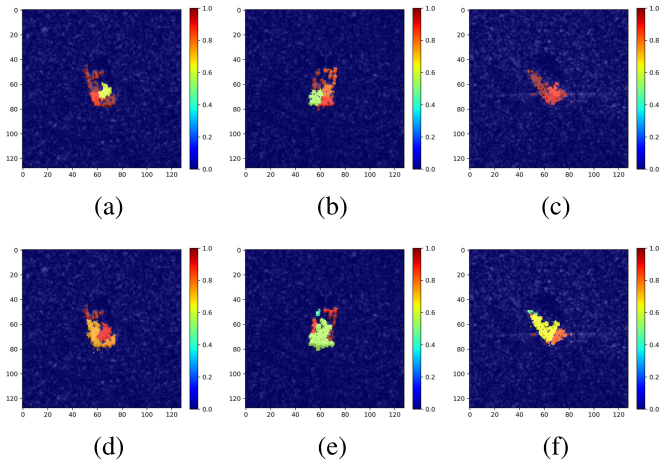



Fig. 18. Part importance maps of vehicle targets. (a)–(c) Targets of correct classification in Kmeans-PAN. (d)–(f) Same targets but misclassified in SC-PAN.

vehicle targets of Kmeans-PAN and SC-PAN under EOC1 with angle normalization are shown in Figs. 16 and 17, respectively. Since Kmeans-PAN has better classification performance than SC-PAN under EOC1, we can see from the figures that the comparison between different parts in the average part importance images of the targets with Kmeans-PAN is more obvious, and the parts that play a leading role in the target classification result are more prominent.

To further investigate the impact of different parts on the target classification results, we visualize the part importance of samples correctly classified in Kmeans-PAN but incorrectly classified in SC-PAN. The part importance maps of these targets are shown in Fig. 18. From the distribution of target parts, the results of part extraction with different algorithms conform to the analysis in Section II-A, that is, the parts obtained by K-means are more evenly divided, and the SC tends to split target into the main body and several smaller parts. Since EOC1 refers to the large depression angle difference between test data and training data, the electromagnetic scattering characteristics of some large structures in the target will change significantly. It can also be seen from Fig. 18 that

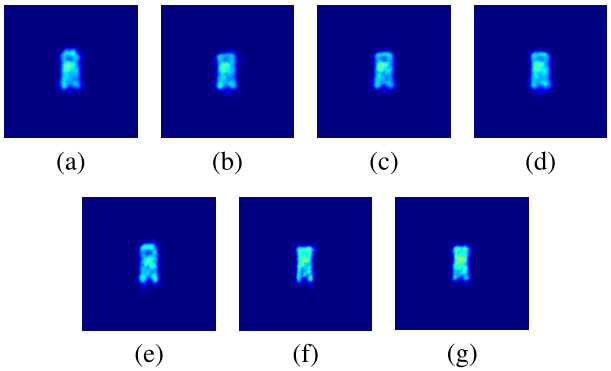



Fig. 19. Average part importance images of MSTAR vehicle targets of Kmeans-PAN under EOC2/V with angle normalization. (a)–(e) T-72 targets with different serial numbers. (f) and (g) BMP-2 targets with different serial numbers.

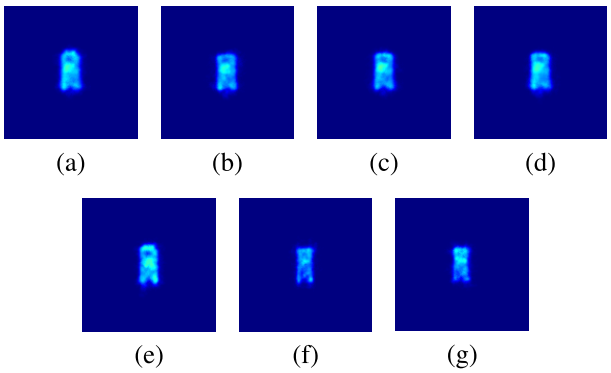



Fig. 20. Average part importance images of MSTAR vehicle targets of SC-PAN under EOC2/V with angle normalization. (a)–(e) T-72 targets with different serial numbers. (f) and (g) BMP-2 targets with different serial numbers.

the larger parts obtained by the SC are of lower importance; however, when these targets are correctly classified, some local structures within the larger parts are actually of higher importance. Therefore, we infer that, because there are usually large parts in the result of SC-based parts segmentation, it is difficult to ensure that the characteristics of test data and training data are consistent under EOC1, resulting in a low recognition rate.

*3) Results of EOC2 Data:* The average part importance images of MSTAR vehicle targets of Kmeans-PAN and SC-PAN under EOC2/V with angle normalization are shown in Figs. 19 and 20, respectively. From the figures, it can be seen that the importance of the parts calculated based on the two algorithms is the same for T-72 targets with different serial numbers, but there is a significant difference for BMP-2 targets. In SC-PAN, the difference between BMP-2 and T-72 is more obvious, which is more conducive to distinguishing.

Tables XV and XVI are the confusion matrices obtained by the two algorithms. By comparing the two tables, we can see that SC-PAN is more advantageous in distinguishing BMP-2 from T-72 than Kmeans-PAN. This result is consistent with the analysis of the parts' importance images. Combining the above results, we infer that, in EOC2 classification tasks, larger parts in the target are usually used to reflect the differences between categories, while some detail parts are

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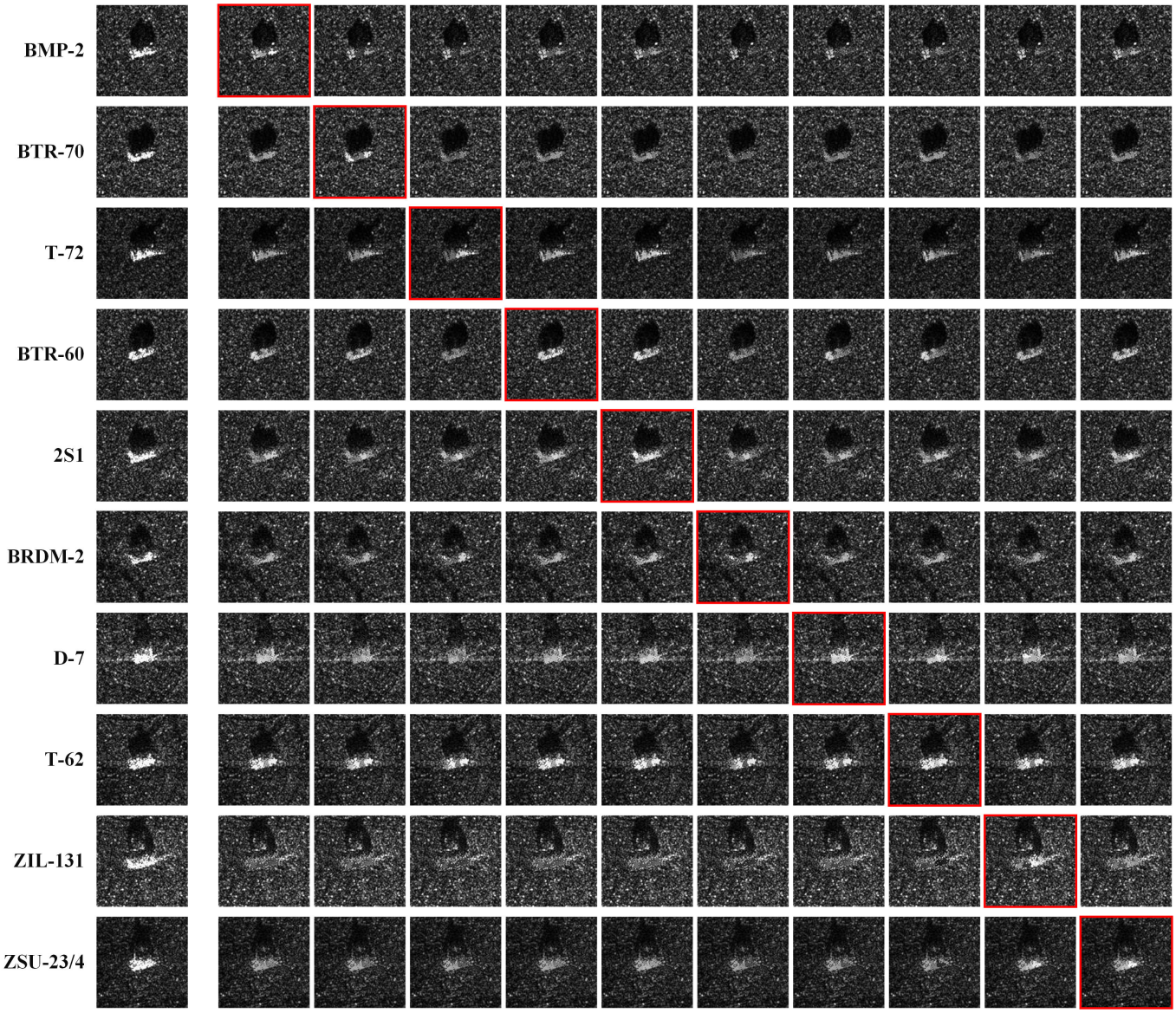



Fig. 21. Original SAR image and decision maps for all ten categories of targets with correct classification results using the proposed deep learning model.

TABLE XV
RECOGNITION RESULTS OF KMEANS-PAN UNDER EOC2/V

| Class/Series | Results | | | |
|---|---|---|---|---|
| | BMP-2 | BTR-70 | T-72 | BRDM-2 |
| BMP-2/9566 | **407** | 9 | 12 | 0 |
| BMP-2/C21 | **409** | 0 | 20 | 0 |
| T-72/A04 | 1 | 0 | **571** | 1 |
| T-72/A05 | 0 | 0 | **572** | 1 |
| T-72/A07 | 0 | 0 | **573** | 0 |
| T-72/A10 | 0 | 0 | **567** | 0 |
| T-72/812 | 5 | 0 | **421** | 0 |

TABLE XVI
RECOGNITION RESULTS OF SC-PAN UNDER EOC2/V

| Class/Series | Results | | | |
|---|---|---|---|---|
| | BMP-2 | BTR-70 | T-72 | BRDM-2 |
| BMP-2/9566 | **412** | 8 | 8 | 0 |
| BMP-2/C21 | **421** | 2 | 6 | 0 |
| T-72/A04 | 1 | 0 | **570** | 2 |
| T-72/A05 | 0 | 0 | **572** | 1 |
| T-72/A07 | 0 | 0 | **573** | 0 |
| T-72/A10 | 1 | 0 | **566** | 0 |
| T-72/812 | 2 | 0 | **424** | 0 |

used to ensure the consistency of different series targets of the same type. Therefore, SC-PAN can achieve better classification performance.

### *E. Decision Maps for SOC Target Images*

In Section III-C, we analyzed the target structures that play a key role in the classification process mainly through the visualization of the part importance map. The conclusion is statistically significant because it is based primarily on the average of the calculated results of part importance for the same category targets. In order to further analyze the role of pixels or structures in the classification process of each target, taking data and experiments under SOC as an example, this section calculates and visualizes the decision basis of each

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target in the deep learning model based on the part importance analysis.

Also, we first calculate the importance of each part of the target to the classification results: $e^{\mathrm{Conv}_k(\mathbf{V_p})}$. Unlike the part importance image, which only calculates the importance for the real category of the target, here, we calculate the importance of all the target parts for all the decision results of the DLN, that is, all the possible categories of the target. Assuming that the targets have $C$ categories and each sample is divided into $n$ parts, from the above calculation, we can get a matrix of $n \times C$ dimension that represents the importance of the parts relative to the network decision results. Then, we multiply the $n$ values of the $k$th category column of the matrix by the pixel values of the corresponding $n$ parts regions in the original SAR image of the target to obtain a decision map where the network model classifies the target into class $k$. Similarly, we can get classification decision maps for all $C$ categories of targets. For the $C$ decision maps of each target, their meanings are actually the visualization of the class probability vectors obtained by the proposed classification network for that target. This method converts a 1-D network output into a 2-D image that provides an intuitive understanding of the basis for network discrimination.

Taking data and experiments under SOC as an example, Fig. 21 shows the original SAR image and decision maps for all ten categories of targets with correct classification results using the proposed Kmeans-PAN. The first column in the figure represents the original SAR image of the sample, and the last ten columns are decision maps of the network's ten possible class prediction outcomes. The map marked with a red box is the decision map of the sample under the correct category label. Compared with other decision maps of wrong category labels, we can find that the pixels in the target area are more obvious, and some structures are more prominent. This indicates that the correct classification of the network depends mainly on these brighter target structures. Furthermore, a high level of consistency can be found by comparing the decision images under these correct categories with the average part importance maps in SOC. In addition, compared with the original images, we can see that most of the correctly classified decision maps only highlight local target structures. This indicates that, in SAR ATR, not all target structures are required, and some key local structures control the target classification results.

## IV. Conclusion

Considering the advantages of both traditional ML and deep learning methods in SAR ATR, a novel PAN integrating electromagnetic scattering characteristics is proposed in this article. First, considering the importance of scattering structure for SAR ATR, we design a target part model based on the ASC model and discuss the effects of different clustering methods on the part results. Then, a PAM based on an attention mechanism is proposed, which effectively fuses the features of target parts and can directly reflect the role of each part in the classification results. Finally, the importance of each part can be deduced based on the PAM combined with the network forward calculation method, which is important for practical application and provides new ideas for the interpretation of SAR ATR. Experiments on the MSTAR dataset demonstrate the effectiveness of the proposed PAN. In addition, this article proposes two interpretable methods to analyze the classification results of the proposed DLN: part importance images and decision maps. By combining visual analysis of the importance of parts to classification results under different experimental conditions, we provide a rational explanation for the deep learning SAR ATR mechanism, especially for misclassified samples. By calculating decision maps, we can further analyze the key structure of each sample in the classification process. From the above work, we can find the target parts that play a key role in the classification task. This will be helpful for future studies, such as simulation based on key parts, adversarial examples generation, and identification of occluded samples.

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